SINDH MADRESSATUL ISLAM UNIVERSITY KARACHI

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LAB 02

ADVANCE STATISTICS

LAB INSTRUCTOR: MISS AQSA

Course: Advance Statistics

Département: Data Science

Semester:4

Lab 2 – Variance & Standard Deviation

Objective:

- Understand the concepts of Variance and Standard Deviation.
- Learn how they measure the spread (dispersion) of data.
- Practice calculating them in Excel / Google Sheets.

Short Theory:

• Deviation = difference between each data point and the mean.
• Variance (σ^2) = average of squared deviations.
  Formula: Variance = $\Sigma (x - \bar{x})^2 / n$
• Standard Deviation (σ) = square root of variance.
  Formula: SD = $\sqrt{\text{Variance}}$
• Variance is in squared units, while Standard Deviation brings it back to the original scale.

Procedure (Excel / Google Sheets):

1. Step 1 – Enter the following dataset (Student Marks):

Students	Marks
1	12
2	18
3	25
4	20
5	30
6	22
7	15

2. Step 2 – Calculate Mean:
   Formula:
   =AVERAGE(B2:B8)
3. Step 3 – Calculate Deviation:
   Formula (for first student):
   =B2 - B9
   (where B9 contains the mean).
4. Step 4 – Square the Deviations:
   Formula:
   =(C2)^2

5. Step 5 – Find Variance:
   Formula:
   =AVERAGE(D2:D8)
6. Step 6 – Find Standard Deviation:
   Formula:
   =SQRT(B10)

   OR directly use Excel's built-in functions:
   - Variance → =VAR.S(B2:B8)
   - Standard Deviation → =STDEV.S(B2:B8)

Student Task:

1. Enter the daily sales data for 10 days.

Day	Sales
1	120
2	135
3	128
4	140
5	150
6	132
7	145
8	138
9	142
10	155

2. Calculate:
   - Mean
   - Variance
   - Standard Deviation
3. Draw a bar chart of the squared deviations.
4. Submit the screenshot of Excel file.